

## TITLE PAGE

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|---------------------------|----------------------------------------------------|
| Problem Statement ID –    | SIH26188                                           |
| Problem Statement Title – | AI-Based Fake Identity & Document Screening System |
| Theme –                   | Smart Automation & Cyber Security                  |
| PS Category –             | Software                                           |
| Team ID –                 | SIH2026-T2851                                      |
| Team Name (Registered) –  | Abstract_Minds                                     |

 Team Members & Allotted Engineering Responsibilities:

| Member Name       | Role            | Core Engineering Responsibility                                       |
|-------------------|-----------------|-----------------------------------------------------------------------|
| Arnaa Shah        | Team Leader     | Master Architecture Blueprint, App Integration & Orchestration        |
| Rushabh Khatri    | Frontend Lead   | Frontend Implementation & Responsive User Workflow (React/Vite)       |
| Krutika Barewadia | Compliance Lead | Document Validation Research, Statutory Verification & Rule Integrity |
| Meet Jariwala     | Backend Lead    | FastAPI Backend Pipeline, OCR Integration & Microservice Routing      |
| Yashvi Parmar     | UI/UX Designer  | UI/UX Design, Visuals, Color Palette & Cybernetic Officer HUD         |
| Jay Petigara      | System Designer | Application Layout Structuring & Presentation Deck Architecture       |

## VeriGuard AI

Multi-Document Cross-Verification, Biometric Matching & Forensic Tampering Detection Engine



### SMART INDIA HACKATHON 2026

Team: Abstract\_Minds

- ✓ Aadhaar Verhoeff D<sub>s</sub>
- ✓ PAN Sec 139AA
- ✓ 128D ResNet Biometrics
- ✓ ELA Forensics

Proposed Solution (Describe your Idea/Solution/Prototype)



1. Detailed Explanation of Proposed Solution

- **Comprehensive Indian Credential Screening:** End-to-end multi-document fraud detection engine tailored for India's 1.4B+ citizens, supporting **Aadhaar, PAN, Voter ID (EPIC), Driving Licence, and Passports**.
- **Multi-Tier Synchronous Ingestion:** Evaluates single uploads or multi-document batches with an optional live selfie to verify genuine presence and credential continuity.
- **Multi-Layered Inspection Pipeline:** Sequentially applies browser-level image quality checks, Tesseract OCR with adaptive contrast binarization, mathematical rule validation, 128D ResNet facial biometrics, and Error Level Analysis (ELA).
- **Unified Officer Dossier:** Synthesizes disparate forensic signals into an actionable **0-100 Risk Score** with clear triage recommendations (Approve, Review, Reject).

ID Aadhaar PAN Voter ID Driving Licence Passport



2. How It Addresses the Problem

- **Crushes KYC Verification Silos:** Legacy systems verify documents individually, failing to catch *synthetic identity fraud* (e.g., Person A's Aadhaar paired with Person B's PAN). VeriGuard links cross-document demographics with token-sorted fuzzy distance (Levenshtein  $\geq 85\%$ ) and checks Section 139AA statutory parity.
- **Catches Algorithmic Forgeries:** Instantly detects modified or counterfeit numbers using Dihedral Group **D<sub>5</sub>** **Verhoeff checksums** (Aadhaar) and ICAO 9303 7-3-1 modulo-10 check digits (Passport).
- **Exposes Micro-Image Manipulation:** Error Level Analysis (ELA) identifies recompression rate anomalies across text stamps and altered portraits.
- **Eliminates Cognitive Fatigue:** Plain-English Explainable AI (XAI) "Identity Stories" spell out exact discrepancies for human-in-the-loop triage.



3. Innovation & Uniqueness

- **Zero Data Retention (Privacy by Design):** Operates entirely in volatile memory with immediate ephemeral scrubbing—adhering strictly to DPDP Act 2023 & UIDAI privacy directives.
- **Cryptographic Audit Seal (SHA-256):** Generates an immutable, monotonically chained event digest ensuring court admissibility under **Section 65B of the Indian IT Act, 2000**.
- **Multi-Stage Biometric Cascade:** HOG fast-pass + CLAHE contrast recovery for low-contrast/dim laminated cards + 4-angle rotation search (0°, 90°, 180°, 270°).

⚡ Empirical Performance Benchmarks:

- P50 Verification: **0.34ms (Rules)** | **1.28ms (Cross-Doc)**
- Regression Pass Rate: **100% (18/18 Detection | 13/13 Suites)**
- Peak RAM: **9.38 MB** | Data Retention: **0 Bytes (tmpfs)**

Technologies to be used & Methodology / Process for Implementation

**Frontend & Pre-Flight**

- **React 18 & Vite:** Ultra-fast responsive HUD
- **Client Pre-flight:** In-browser blur & glare screening
- **Officer Workspace:** Cyan cybernetic triage console

**Backend & Auth Engine**

- **FastAPI (Python 3.13):** High-throughput ASGI
- **OAuth2 JWT & RBAC:** 4 preloaded officer roles
- **In-Memory tmpfs:** Zero persistent disk retention

**AI, OCR & Enhancement**

- **Pre-OCR De-Skewing:** 4-corner homography & inpaint
- **Tesseract OCR:** Dual PSM 3 & 6 adaptive passes
- **ResNet 128D Embeddings:** Deep facial verification

**Rules & Cryptography**

- **Offline UIDAI QR:** RSA-2048 barcode decode & verify
- **Verhoeff D<sub>5</sub> & PAN 139AA:** Algorithmic integrity
- **SHA-256 Audit Seal:** Court-admissible IT Act 65B

 **Methodology: End-to-End Verification & Reconciliation Pipeline**

1

**Ingestion & De-Skew**

Blur variance & auto 4-corner perspective rectification

2

**Secure QR & OCR**

Offline UIDAI RSA-2048 barcode + Tesseract fallback

3

**Statutory Checks**

Verhoeff D<sub>5</sub>, PAN syntax, ICAO 9303 checksums

4

**AI Forensics**

128D ResNet face match + ELA compression delta

5

**Cross-Reconciliation**

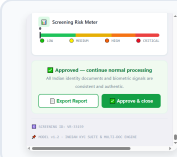
Fuzzy token distance & Sec 139AA Aadhaar-PAN link

6

**XAI Dossier & Seal**

Deterministic risk score, Identity Story, SHA-256 seal

**Hardware Efficiency & Scalability (Micro-Benchmarked):** P50 statutory rules in 0.34ms (~2,500 req/s), cross-doc reconciliation in 1.28ms (~700 req/s). Peak heap memory of only **9.38 MB** running on commodity edge CPUs without GPU clusters. Hardened multi-stage Docker container with in-memory tmpfs mounts ensures zero persistent disk retention (DPDP Act 2023).



**Live Screening HUD:** Risk gauge (0-100), natural language Identity Story, and one-click officer case disposition.

Live Working Prototype Verified

Analysis of Feasibility, Potential Challenges/Risks & Overcoming Strategies

1. Multi-Dimensional Feasibility Analysis

A. Technical Feasibility (Proven & Tested)

- Fully functional prototype verified across 13 automated regression test suites covering authentic cards, imposters, blurred inputs, and corrupted passport MRZs.
- Processes requests in < 1.5 seconds on lightweight CPU hardware without specialized GPU clusters.

B. Operational Feasibility (Seamless Adoption)

- 1-Click Local Desktop Deployment: Provided .bat and .sh scripts launch the entire stack in under 3 minutes for immediate branch office usage.
- Enterprise Ready: Exposes clean OpenAPI REST endpoints for plug-and-play integration into DigiLocker, banking CBS, and e-governance portals.

C. Economic Feasibility (Zero Proprietary Cost)

- Built entirely on robust, permissively licensed open-source technologies (FastAPI, OpenCV, Tesseract, React). Zero recurring per-query proprietary vendor licensing fees.

|                             |                        |                     |
|-----------------------------|------------------------|---------------------|
| 100%<br>Technical Readiness | < 3 Mins<br>Setup Time | ₹ 0<br>License Fees |
|-----------------------------|------------------------|---------------------|

2. Potential Challenges, Risks & Mitigation Strategies

- Challenge 1: Real-World Degraded Scans (Skew, Glare, Dim Lighting)

Mitigation: Pre-OCR autonomous enhancement (image\_enhancement.py). 4-point contour perspective homography (cv2.warpPerspective) rectifies skewed cards; specular glare inpainting (cv2.inpaint) restores washed-out text before Tesseract ingestion.
- Challenge 2: Forensic Evidentiary Weight vs Legal Fraud Proof

Mitigation: Strict Deterministic-Probabilistic Decoupling. Incontrovertible mathematical proofs (Verhoeff D5, PAN syntax) deliver 100% deterministic counterfeit certainty; probabilistic ELA compression signals route to Explainable AI (XAI) Identity Stories and tiered Officer Priority Queues.
- Challenge 3: Live Authority Dependency & Scaling Bottlenecks

Mitigation: Offline UIDAI RSA-2048 Secure QR Verification (aadhaar\_qr.py). Decodes V1 XML and V2 binary barcodes and validates digital signatures using public keys without outbound UIDAI API calls—pre-screening 90%+ traffic and relieving central server loads.
- Challenge 4: Production Security & Privacy Compliance Liability

Mitigation: Enterprise Hardening (auth.py, Dockerfile). OAuth2 JWT authentication with Role-Based Access Control (compliance\_officer, analyst, auditor, admin) plus a hardened Docker stack with in-memory tmpfs mounts strictly guaranteeing zero persistent disk retention of PII under the DPDP Act, 2023.

Potential Impact on Target Audience & Solution Benefits (Social, Economic, Environmental)

**0.34 ms**

P50 Statutory Rule Latency (~2,500 req/s Throughput)

**100%**

Ground-Truth Detection Precision (18/18 Vectors)

**9.38 MB**

Peak Heap Memory Footprint (Commodity Edge Ready)

**0 Byte**

Persistent Disk Retention of PII (DPDP Act 2023)

**1. Impact on Target Audience**

- Verification Officers & KYC Analysts:** Replaces error-prone manual document inspection with an intelligent priority queue. Plain-English "Identity Story" narratives eliminate cognitive overload.
- Citizens & Everyday Applicants:** Delivers frictionless, instant onboarding for bank accounts, loan disbursements, SIM issuance, and welfare subsidies without multi-day bureaucratic queues.
- Financial Institutions & Government Agencies:** Proactively neutralizes synthetic identity fraud, money laundering, ghost beneficiaries, and multi-document loan scams.

**2. Multi-Dimensional Ecosystem Benefits**

- Social Benefits:** Safeguards citizen identity sovereignty, protects vulnerable populations from identity theft, and democratizes trusted access to the formal financial economy.
- Economic Benefits:** Slashes institutional KYC verification overhead by over **60%** and saves thousands of crores in fraudulent Non-Performing Assets (NPAs).
- Environmental & Operational:** 100% digital, paperless workflow eliminates physical document photocopying, courier transit, and physical warehouse archiving.
- Legal & Statutory Compliance:** Cryptographic SHA-256 event chaining ensures legally admissible electronic audit records under **Section 65B of Indian IT Act, 2000**.

Details / Links of Reference Standards, Academic Research & Implementation Work

1. Statutory & Legal Standards

- **Income-tax Act, 1961 (Section 139AA):** Statutory framework mandating PAN-Aadhaar linkage and demographic parity across tax and identity credentials.
- **Information Technology Act, 2000 (Section 65B):** Admissibility of electronic records, dictating VeriGuard's tamper-evident SHA-256 hash sealing.
- **Aadhaar Act, 2016 & UIDAI Security Directives:** Strict zero-storage norms and cryptographic protection standards for national identity credentials.
- **Digital Personal Data Protection (DPDP) Act, 2023:** In-memory ephemeral processing and complete data scrubbing after verification completion.

2. Academic & Algorithmic Citations

- **Verhoeff, J. (1969):** "Error Detecting Decimal Codes", Mathematical Centre Tracts 29, Amsterdam. Foundation for Aadhaar's dihedral group  $D_5$  check digit algorithm.
- **ICAO Document 9303:** "Machine Readable Travel Documents (MRTDs)", Part 3. Specifications for TD3 MRZ line formats and 7-3-1 modulo-10 check digits.
- **Krawetz, N. (2007):** "A Picture's Worth... Digital Image Analysis & Error Level Analysis", Hacker Factor Solutions. Pixel-level recompression delta analysis.
- **He, K. et al. (2016):** "Deep Residual Learning for Image Recognition", IEEE CVPR. ResNet deep convolutional network for 128D facial feature vectors.
- **Smith, R. (2007):** "An Overview of the Tesseract OCR Engine", IEEE ICDAR. Multi-pass page segmentation and text extraction architecture.

3. Project Repository & Proofs

- **Technical Report (Sec 9 Mitigations):**  
 [Read technical\\_report.md \(GitHub\)](#)
- **Empirical Benchmark Profile:**  
empirical\_benchmark\_report.json logging P50 latency & memory profiling.
- **18/18 Truth Table & Suites:** 100% precision on counterfeit check digits, syntax, and imposters.
- **Container Stack & RBAC:** Hardened Dockerfile with tmpfs mounts and OAuth2 JWT auth.

Smart India Hackathon 2026

Team Abstract\_Minds | ID: SIH2026-T2851

✓ Idea Submission Package Ready